Daily Problem: 15–Feb–2026

Problem Statement

Prove that no weighted automaton over the real numbers with singleton alphabet $\{a\}$ can express the function

$$f(a^n) \;=\; e^{-\lambda}\,\frac{\lambda^n}{n!} \qquad (n \geq 0),$$

i.e. the Poisson distribution with parameter $\lambda > 0$.

Model of a weighted automaton over $\mathbb{R}$

A (real) weighted automaton over $\{a\}$ is a tuple

$$A = (Q, \{a\}, \mu, \alpha, \beta),$$

where Q is a finite set of states, $\mu(a) \in \mathbb{R}^{|Q|\times|Q|}$ is the transition matrix, $\alpha \in \mathbb{R}^{1\times|Q|}$ is the initial row vector, and $\beta \in \mathbb{R}^{|Q|\times 1}$ is the final column vector. It assigns to each word a^n the weight

$$g(a^n) \;=\; \alpha\,\mu(a)^n\,\beta \qquad (n \geq 0).$$

Thus any unary weighted automaton defines a real sequence $(g_n)_{n\geq 0}$ where

$$g_n := g(a^n) = \alpha M^n \beta, \qquad M := \mu(a).$$

Key fact: unary weighted automata yield rational generating functions

Let $m = |Q|$. Consider the ordinary generating function

$$G(z) \;=\; \sum_{n=0}^{\infty} g_n z^n.$$

For any weighted automaton A with m states, the series $G(z)$ is a rational function of z.

Proof. We use the identity (formal power series / matrix resolvent)

$$\sum_{n=0}^{\infty} (zM)^n \;=\; (I - zM)^{-1}.$$

Multiplying on the left by α and on the right by β gives

$$G(z) = \sum_{n=0}^{\infty} \alpha (zM)^n \beta = \alpha \left(\sum_{n=0}^{\infty} (zM)^n \right) \beta = \alpha (I - zM)^{-1} \beta.$$

Each entry of $(I - zM)^{-1}$ is a ratio of polynomials in z (it is $\operatorname{adj}(I - zM)/\det(I - zM)$). Hence $G(z)$ is a rational function. □

So, *if* a unary weighted automaton represents the desired function f, then its generating function must be rational.

Poisson weights have a non-rational generating function

Now consider the target sequence

$$f_n := f(a^n) = e^{-\lambda} \frac{\lambda^n}{n!}.$$

Its ordinary generating function is

$$F(z) := \sum_{n=0}^{\infty} f_n z^n = e^{-\lambda} \sum_{n=0}^{\infty} \frac{(\lambda z)^n}{n!} = e^{-\lambda} e^{\lambda z}.$$

If $\lambda > 0$, then $e^{\lambda z}$ is not a rational function of z.

Proof. Suppose for contradiction that $e^{\lambda z}$ is rational. Since $e^{\lambda z}$ is an entire function (analytic on all of $\mathbb{C}$), it has no poles. But any nonconstant rational function has a pole somewhere in $\mathbb{C} \cup \{\infty\}$; equivalently, a rational function that is entire must be a polynomial.

Therefore $e^{\lambda z}$ would have to be a polynomial.

But $e^{\lambda z}$ is not a polynomial: it has infinitely many nonzero Taylor coefficients at 0,

$$e^{\lambda z} = \sum_{n=0}^{\infty} \frac{\lambda^n}{n!} z^n,$$

where $\frac{\lambda^n}{n!} \neq 0$ for all n. A polynomial has only finitely many nonzero Taylor coefficients. This contradiction shows $e^{\lambda z}$ is not rational. □

Since $F(z) = e^{-\lambda} e^{\lambda z}$, multiplying by the nonzero constant $e^{-\lambda}$ does not change rationality. Hence $F(z)$ is not rational.

Final contradiction

Assume, for contradiction, that there exists a weighted automaton over $\mathbb{R}$ on alphabet $\{a\}$ that expresses the Poisson function, i.e. $g(a^n) = f(a^n)$ for all $n \geq 0$.

Then the corresponding generating functions must agree:

$$G(z) = F(z).$$

By Lemma , $G(z)$ is rational. But by Lemma , $F(z) = e^{-\lambda} e^{\lambda z}$ is not rational. Contradiction.

Therefore, no weighted automaton over the reals with singleton alphabet $\{a\}$ can express $f(a^n) = e^{-\lambda} \frac{\lambda^n}{n!}$.

Conclusion

Unary weighted automata over $\mathbb{R}$ always define sequences whose ordinary generating function is rational. The Poisson weights have generating function $e^{-\lambda} e^{\lambda z}$, which is not rational for $\lambda > 0$. Hence the Poisson distribution cannot be represented by such a weighted automaton.